

Design and performance evaluation of alkali activated materials produced with multiple wastes

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ABSTRACT

Industrial solid wastes offer compelling precursors for developing low-carbon alkali-activated materials; however, the interactions among multiple waste components in quaternary systems remain insufficiently understood. This study develops and systematically characterizes an alkali-activated binder incorporating ground granulated blast-furnace slag (GGBS), steel slag (SS), limestone powder (LP), and calcium carbide residue (CCR). Binder compositions were optimized through sequential single-factor and orthogonal design, and their fresh properties, mechanical performance, and microstructural evolution were evaluated. The results indicate that system performance depends on the combined effects of GGBS reactivity, CCR-mediated activation, LP-induced particle packing and nucleation, and SS incorporation. The optimal formulation (49% GGBS, 25% SS, 20% LP, 6% CCR) achieved a flow diameter of 256 mm, initial and final setting times of 14.1 and 19.5 min, zero 2-h bleeding, and a 28-day compressive strength of 75.8 MPa. Microstructural analysis suggested that this performance was associated with the formation of poorly crystalline C-(A)-S-H gels, pore refinement, and improved particle-matrix interfaces. In contrast, increasing the SS contents while reducing the GGBS content results in later-age strength regression accompanied by residual crystalline phases, the persistence of high-calcium phases, increased accessible pore volume, and interfacial debonding around unreacted particles, collectively limiting the contribution of newly formed reaction products to the load-bearing matrix. These findings indicate that favorable performance in quaternary systems depends on maintaining a balanced precursor composition that aligns reactive supply, alkaline activation, and structural stability. The findings provide experimental and mechanistic evidence for rational design of high-volume solid waste-based alkali-activated materials.

1. Introduction

The construction industry faces increasing pressure to reduce the environmental impact associated with ordinary Portland cement (OPC) production, which is characterized by high energy consumption and substantial CO₂ emissions [1–5]. Consequently, the

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development of sustainable construction materials based on industrial by-products and solid wastes has attracted significant attention. Among the available alternatives, alkali-activated materials (AAMs) have emerged as promising low-carbon binders owing to their ability to incorporate large quantities of industrial waste while achieving excellent mechanical performance and durability [6,7]. AAMs are inorganic binder materials synthesized from aluminosilicate precursors and industrial by-products activated by alkaline solutions [8,9]. Owing to their superior mechanical performance, excellent corrosion resistance, and significantly lower carbon footprint, AAMs have attracted considerable attention as environmentally friendly construction materials [10–13]. The replacement of Portland clinker with reactive industrial by-products therefore represents an effective strategy for reducing the environmental burden of cement-based materials while maintaining or even enhancing engineering performance [14–18].

Multi-source industrial solid wastes rich in Ca–Si–Al components represent promising precursor materials for developing low-carbon alkali-activated materials with reduced clinker dependence [19]. Among these, alkali-activated materials (AAMs) have attracted sustained research interest owing to their rapid early-age reactivity, high strength-development potential, and broad compatibility with diverse solid-waste feedstocks [20,21]. Ground granulated blast-furnace slag (GGBS), a by-product of the iron-making industry, possesses a fineness comparable to that of ordinary Portland cement (OPC) [22]. Owing to its latent hydraulic nature, GGBS can undergo hydration to form calcium silicate hydrate (C–S–H) gel, and this reaction is significantly enhanced in the presence of alkaline activators such as NaOH, $\text{Ca}(\text{OH})_2$, Na_2SiO_3 , and Na_2CO_3 [23,24], providing the principal source of mechanical strength and structural continuity [25]. Mineral fillers such as limestone powder (LP) further enhance fresh-paste rheology, refine pore structure, and mitigate shrinkage through combined particle-filling, heterogeneous nucleation, and packing-adjustment effects [26]. These characteristics highlight that performance optimization of solid-waste-based alkali-activated materials require coordinated control of reaction kinetics, fresh-state workability, and hardened microstructure through deliberate multi-component design.

In multi-solid-waste systems, different components may provide complementary functions. Reactive aluminosilicate precursors contribute to hydration-product formation, mineral fillers improve matrix compactness, and alkaline calcium sources promote precursor dissolution and early-age reactions [27]. Compared with conventional cementitious materials, alkali-activated materials offer significantly lower production costs and a more streamlined manufacturing process [27–29]. You and Gomez-Casero [30,31] reported that alkali activation effectively imparts cementitious properties to slag, resulting in alkali-activated composites with excellent mechanical performance. Limestone powder, in particular, is not merely an inert diluent in alkali-activated systems — its combined filling, nucleation, and dilution effects modulate hydration-product formation, pore-structure development, and engineering properties across a range of binder compositions [32,33]. Calcium carbide residue (CCR), which is predominantly composed of $\text{Ca}(\text{OH})_2$, serves as a low-cost internal alkaline activator and supplementary calcium source, supporting alkali activation without dependence on externally applied high-purity chemical activators [34,35]. Accordingly, designing binder systems based on the complementary roles of individual wastes — in physical filling, chemical activation, and reactive-phase supply — provides an effective strategy for simultaneously improving solid-waste utilization efficiency and engineering performance.

However, multi-waste synergy is not equivalent to the simple physical blending of multiple components. Different industrial by-products differ substantially in reactivity, alkali demand, particle morphology, and volume stability; when proportions are not properly balanced, continued hydration does not necessarily translate into the formation of a stable and continuous load-bearing matrix. Steel slag (SS) is particularly complex in this regard. It contains potentially hydraulic calcium silicate phases — including dicalcium silicate (C_2S) and, in some cases, tricalcium silicate (C_3S) — that may partially hydrate in alkaline environments. At the same time, low-reactivity or expansive residual phases, including the RO phase, free CaO (f-CaO), and free MgO, are prone to delayed hydration or phase transformation at later ages, potentially inducing local volumetric instability and compromising the structural integrity of the hardened matrix [36,37]. When SS is incorporated into alkali-activated binder systems, its potential cementitious contribution must therefore be carefully balanced against the risk of later-age structural deterioration [38–40].

Research on industrial solid-waste-based alkali-activated materials has predominantly focused on single-waste utilization or binary and ternary systems [41], with limited understanding of multi-component synergistic interactions in higher-order systems [42]. The GGBS–SS–LP–CCR quaternary system investigated in this work is designed on the basis of functional complementarity: GGBS serves as the primary reactive Ca–Si–Al source; LP improves particle packing and provides heterogeneous nucleation sites; CCR supplies internal alkalinity and supplementary calcium; and SS contributes potential cementitious reactivity while also introducing residual-phase risks at later ages. The central scientific question addressed here is whether these multi-source solid wastes can be proportioned to simultaneously achieve favorable fresh-state workability, high mechanical performance, and a stable, continuously evolving microstructure. In this study, the term “synergistic effect” refers to the complementary interaction among GGBS, SS, LP, and CCR, whereby their combined contributions to hydration, particle packing, and microstructural development result in superior engineering performance compared with the individual effects of each component.

To address these research gaps, this study developed a GGBS–SS–LP–CCR quaternary alkali-activated binder based on the functional complementarity of multiple industrial solid wastes. Single-factor experiments were first conducted to evaluate the effects of LP, SS, and CCR on fresh-state and mechanical properties, followed by an $\text{L}_9(3^4)$ orthogonal design for mixture optimization. The optimal mixture (Z6) and the mixture exhibiting later-age strength regression (Z8) were selected for detailed microstructural investigation, with LP-only and SS-only binary mixtures serving as reference groups. X-ray diffraction (XRD), thermogravimetric analysis coupled with differential scanning calorimetry (TG–DTG), mercury intrusion porosimetry (MIP), and scanning electron microscopy with energy-dispersive X-ray spectroscopy (SEM–EDS) were employed to correlate macroscopic performance with phase evolution, hydration-product formation, pore-structure development, and fracture-surface morphology. The study aims to clarify the combined activation mechanism and microstructural evolution governing the GGBS–SS–LP–CCR system, and to provide experimental evidence for the rational design and microstructural optimization of high-volume industrial solid waste-based alkali activated materials.

While GGBS, SS, LP, and CCR have each been individually studied in alkali-activated systems, the present study advances beyond

existing binary and ternary investigations in three respects. First, we identify a compositional threshold in the quaternary GGBS-SS-LP-CCR system beyond which the material transitions from synergistic densification to structural instability — a finding that cannot be captured in simpler binary or ternary formulations. Second, we provide direct microstructural evidence, through combined MIP and SEM-EDS analysis, demonstrating that continued hydration-product formation does not guarantee matrix densification; in high-SS systems, the ink-bottle effect and interfacial debonding around unreacted SS particles prevent newly formed products from contributing to a load-bearing structure. Third, we propose a four-mechanism coordination framework — reactive-phase supply (GGBS), internal alkaline activation (CCR), filling and nucleation (LP), and structural stability risk (SS) — that provides a transferable design principle for other multi-waste systems beyond the specific materials studied here.

2. Materials and methodology

2.1. Raw materials

Ground granulated blast-furnace slag (GGBS), steel slag (SS), limestone powder (LP), and calcium carbide residue (CCR) were used as the constituent solid wastes in this study; all materials were commercially available industrial by-products. The densities of GGBS, LP, SS, and CCR were 3.12, 2.63, 3.74, and 1.75 g/cm³, respectively, while their specific surface areas were 442, 380, 425, and 340 m²/kg, respectively. The median particle diameters (d₅₀) for GGBS, SS, LP, and CCR are roughly 10.32 μm, 20 μm, 80 μm and 30 μm, respectively. Prior to use, all solid materials were dried at 105 °C for 24 h, ground, and sieved through a 200-mesh square-hole sieve, with the retained fraction maintained below 5%. Chemical compositions were determined by X-ray fluorescence spectroscopy, and particle-size distributions were measured by laser diffraction analysis. The results are presented in Table 1 and Fig. 1.

The alkaline activator was prepared by mixing commercial sodium silicate solution (silicate modulus, $M_s = \text{SiO}_2/\text{Na}_2\text{O}$ molar ratio of 3.30) with analytical-grade sodium hydroxide (purity ≥96%). Sodium hydroxide was added to adjust the final activator modulus to 1.6. The solution was then sealed and cooled to room temperature before use. The activator modulus ($\text{SiO}_2/\text{Na}_2\text{O}$ molar ratio) was fixed at 1.6 based on preliminary experiments and previous studies [43,44]. Lower moduli caused rapid setting and reduced workability, while higher moduli decreased activation efficiency and early-age strength. Therefore, a modulus of 1.6 was adopted as the optimum value. The water-to-binder ratio (w/b) reported in this study was calculated based on the total mixing water divided by the total binder mass. The total mixing water included both the free water added during mixing and the water contained in the commercial sodium silicate solution. The sodium silicate solution consisted of 26.98% SiO₂, 8.53% Na₂O, and 64.49% H₂O by mass. The water utilized in the experiment was deionized water.

2.2. Mix proportion

A two-stage experimental strategy was adopted. Single-factor experiments were first conducted to determine the individual effects of LP, SS, and CCR on the fresh and mechanical properties. In these tests, the total binder content was fixed at 100%, and each admixture was introduced by replacing GGBS at equivalent mass fractions. The proportions of LP, SS and CCR in binder materials vary from 10% to 40%, 15% to 45%, and 2% to 14% respectively. All single-factor mixtures were prepared with a fixed alkali content (Na₂O %) of 6%, activator modulus of 1.6 and a water-to-binder ratio (W/B) of 0.4. The detailed mixed proportions are presented in Table 2. The alkali dosage (Na₂O%) was defined as the equivalent Na₂O content expressed as a percentage of the total binder mass. The equivalent Na₂O was supplied by both the commercial sodium silicate solution and sodium hydroxide, while the total binder comprised GGBS, SS, LP, and CCR.

Based on the effective adjustment ranges identified from the single-factor tests, a standard L9(3⁴) orthogonal array was adopted for multi-factor optimization. Solid-waste proportion (Factor A), alkali dosage (Factor B), and W/B ratio (Factor C) were selected as the primary variables, with one empty column reserved for error estimation. CCR content and activator modulus were fixed at 6% and 1.6, respectively, based on preliminary screening and literature guidance. Mix proportions for the orthogonal array are given in Table 3. The CCR content was fixed at 6 wt% based on preliminary experiments. Lower dosages provided insufficient alkalinity, whereas higher dosages accelerated setting and reduced workability. Therefore, 6 wt% was adopted as the optimum dosage.

The influence of each factor on paste performance was evaluated using range analysis. Range R was calculated according to Eq. (1):

$$R = \max(K_1, K_2, K_3) - \min(K_1, K_2, K_3) \quad (1)$$

where K_1 , K_2 , and K_3 represent the mean values of a given performance index at the corresponding levels of each factor. A larger R

Table 1
Chemical compositions of raw materials.

Raw materials	Chemical Composition (wt.%)							
	SiO ₂	Al ₂ O ₃	MgO	CaO	Fe ₂ O ₃	Na ₂ O	K ₂ O	LOI
GGBS	34.28	15.62	8.15	37.54	0.87	0.42	0.58	2.54
LP	1.24	0.35	0.41	53.82	0.18	0.07	0.09	43.84
SS	17.35	4.82	5.74	43.26	15.83	0.13	0.27	2.60
CCR	2.87	0.94	0.56	68.32	0.31	0.08	0.11	26.81

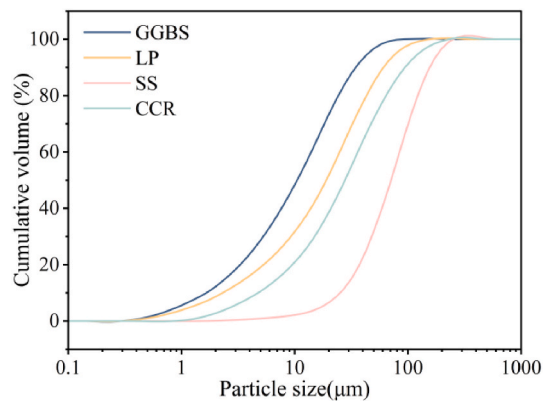


Fig. 1. Particle-size distribution of raw materials.

Table 2

Mix proportions of the single-factor experiments.

Mix ID	GGBS replacement ratio (%)				Ms	Na ₂ O (%)	W/B
LP	10	20	30	40	1.6	6	0.4
SS	15	25	35	45			
CCR	2	6	10	14			

Table 3

Mix proportions of orthogonal experiments.

Mix ID	Binder materials composition (wt.%)				Na ₂ O (%)	W/B
	GGBS	LP	SS	CCR		
Z1	69	10	15	6	4	0.4
Z2					5	0.5
Z3					6	0.6
Z4	49	20	25	6	4	0.5
Z5					5	0.6
Z6					6	0.4
Z7	29	30	35	6	4	0.6
Z8					5	0.4
Z9					6	0.5

value indicates a greater influence of the factor on the evaluated property. Fluidity, setting time, bleeding rate, and compressive strength were selected as performance indices to determine the relative significance of each factor and identify the optimal parameter combination.

It should be acknowledged that the orthogonal design treats the solid-waste proportions (GGBS, LP, SS) as a single combined factor, which does not allow independent quantification of individual component interactions. However, the single-factor experiments independently varied LP, SS, and CCR, enabling isolation of their individual effects. The interaction mechanisms proposed in this study are therefore inferred from the convergence of single-factor trends, orthogonal optimization results, and comparative microstructural analysis of Z6 and Z8. Future studies employing full factorial or response-surface designs would provide more rigorous statistical quantification of interaction effects.

2.3. Sample preparation

The sample preparation methodology is outlined as follows. Initially, the solid raw materials were introduced into the mixing pot and dry-mixed at a slow rate for 90 s until a homogeneous blend was obtained. Subsequently, the alkali activator and additional water were added, and the mixture was gently stirred for 90 s to form a uniform slurry. The material adhering to the bottom of the mixing pot was then scraped using a putty knife for 30 s while the mixer was paused to ensure complete mixing. Finally, the slurry was mixed at high speed for an additional 90 s to achieve full homogeneity. The fresh paste was cast into 40 mm × 40 mm × 160 mm molds in two layers and compacted by vibration for 60 s to remove entrapped air. The surface was leveled and covered with plastic film to prevent moisture loss. Specimens were demolded after 24 h of curing at room temperature and then transferred to a standard curing chamber until the specified testing ages. The curing conditions were 20 ± 2 °C and relative humidity of 95%.

After mechanical testing at the designated ages, fragments (5–10 mm) were collected from interior of fractured specimens and

immediately immersed in anhydrous ethanol for 48 h to terminate hydration. The samples were then dried to constant mass in a vacuum oven at 40 °C. The hardened fragments were used for MIP and SEM–EDS analysis, while a portion of the dried material was ground in an agate mortar and passed through a 200-mesh sieve for XRD and TG–DSC characterization.

2.4. Experimental method

2.4.1. Fresh properties

The fluidity was measured in accordance with GB/T 8077–2012. The average flow diameters were assessed in two perpendicular directions. The setting time was measured using a Vicat apparatus in accordance with GB/T 1346–2011. The initial and final setting times were determined as the mean values of three parallel measurements. The 2-h bleeding rate was determined using the graduated-cylinder method. Fresh paste was poured into a 100 mL graduated cylinder to a volume of 90–100 mL, and the cylinder was immediately sealed with plastic film to prevent evaporation and allowed to stand undisturbed for 2 h. The volume of free water collected at the surface was recorded, and the bleeding rate was calculated as the ratio of the collected water volume to total paste volume. Each mixture was tested in triplicate, and the average value was reported.

2.4.2. Mechanical properties

The mechanical strength tests were conducted in accordance with GB/T 17671–2021 at 7 and 28 days. The loading rates for the mechanical strength tests were set at 50 N/s for flexural strength and 2.4 kN/s for compressive strength. Each specimen was first tested in three-point flexure, and the fractured halves were subsequently used for compressive strength determination. Mean values from three specimens are reported for both indices.

2.4.3. Microstructural analysis

The phase assemblage was identified through the application of X-ray diffraction (XRD), thermogravimetry (TG–DTG), mercury intrusion porosimetry (MIP), and scanning electron microscope (SEM). XRD analysis was performed using the Bruker D8. The scanning range was 10–80° with a step size of 0.02°, and a scanning rate of 2°/min. XRD was used to identify crystalline phases such as calcite, $\text{Ca}(\text{OH})_2$, and steel-slag residual phases. TG–DSC was conducted using the German Netzsch STA449 F3 within a temperature range of 30–1000 °C, with a consistent heating rate of 10 °C/min, and nitrogen was employed as the test gas atmosphere. MIP was performed using a Micromeritics AutoPore V9620. Fractured 5–10 mm samples were dried to constant mass. The pressure range applied during the test was 0.2–300 MPa. SEM–EDS was conducted using a ZEISS GeminiSEM 300. Fracture surfaces were gold-coated and observed at 3 kV. Elemental distributions (O, Na, Mg, Al, Si, Ca, Fe) were analyzed by point and area scanning. SEM was used to relate microstructural features to MIP-derived pore characteristics.

3. Results and discussion

3.1. Fresh properties

The field performance of alkali activated materials is largely determined by their fresh properties, including fluidity, setting time, and bleeding resistance [45,46]. As shown in Figs. 2–4 and Table 4, the fresh-state performance of the paste was jointly controlled by

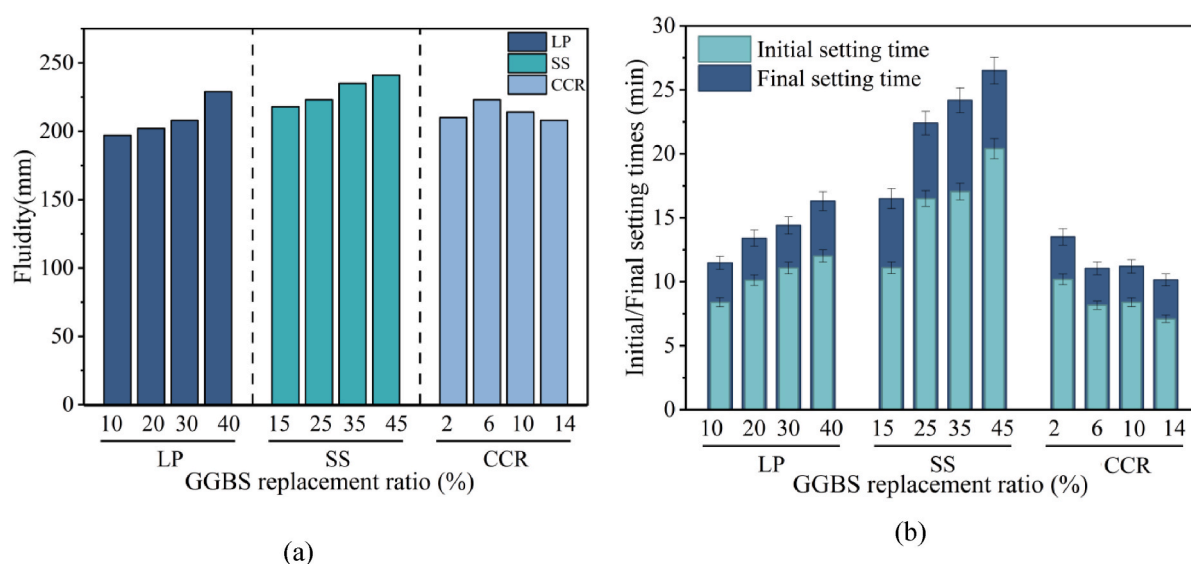


Fig. 2. Fluidity and setting time under different solid-waste contents in the single-factor tests: (a) fluidity; (b) setting times.

LP, SS, and CCR through their respective effects on particle packing, reaction kinetics, and internal alkaline activation [47].

Regarding fluidity, increasing the LP content from 10% to 40% increased the paste flow diameter from 197 mm to 229 mm without noticeable bleeding. This improvement can be attributed to the micro-filling and particle-packing effects of LP. Fine LP particles occupy void spaces between larger particles, resulting in a denser packing structure and reduced interparticle friction during flow [27, 48]. Moreover, LP acts as a relatively low-reactivity component during the early stage, reducing water consumption associated with rapid hydration reactions and maintaining sufficient free water for particle movement. The absence of bleeding further indicates that LP contributed to improved water retention and enhanced stability during static placement. Similarly, increasing the SS content from 15% to 45% increased the fluidity from 218 mm to 241 mm. Compared with GGBS, SS exhibits lower early-age reactivity and slower dissolution kinetics. Consequently, replacing GGBS with SS reduces the consumption of free water caused by early hydration reactions and delays the formation of hydration products, thereby maintaining lower paste viscosity and improving flowability. However, excessive SS may also reduce the amount of reactive phases available for matrix development. In contrast, excessive CCR may negatively affect fluidity. The $\text{Ca}(\text{OH})_2$ present in CCR dissolves rapidly and releases OH^- ions, accelerating precursor dissolution and early hydration reactions. Rapid formation of hydration products increases particle interactions and paste viscosity, thereby reducing fluidity [49].

As for setting behavior, LP slightly prolonged the setting process, with the initial and final setting times increasing from 8.4 to 12.0 min and from 11.5 to 16.3 min, respectively, as the LP content increased from 10% to 40%. This delay is mainly attributed to the dilution effect caused by replacing reactive GGBS with relatively inert LP, resulting in reduced reaction intensity during the early stage. SS exhibited a more pronounced retardation effect on setting. Increasing the SS content from 15% to 45% increased the initial and final setting times from 11.1 to 16.5 min to 20.4 and 26.5 min, respectively. This behavior is associated with the lower early-age reactivity of SS compared with GGBS. The reduction in rapidly dissolving glassy phases decreased the hydration rate and delayed the formation of a continuous hardened structure. Conversely, CCR significantly accelerated setting. As the CCR content increased from 2% to 14%, the initial setting time decreased from 10.22 min to 7.16 min. Dissolution of $\text{Ca}(\text{OH})_2$ released OH^- ions and increased the alkalinity of the system, promoting the depolymerization of the GGBS glass network and accelerating hydration-product formation. Considering the balance between workability and setting behavior, the CCR content was fixed at 6% in the orthogonal experiments.

The effects of the orthogonal factors on the fresh properties are shown in Figs. 3 and 4. The results indicate that the water-to-binder ratio exerted the greatest influence on paste fluidity and bleeding rate, whereas the solid-waste composition had a more significant effect on setting behavior. Mixtures with a higher water-to-binder ratio generally exhibited greater flowability. Increasing the water content reduced particle interactions and lowered paste viscosity, thereby enhancing particle mobility during flow. For example, the mixtures with $W/B = 0.6$ generally exhibited higher fluidity than those with $W/B = 0.4$. However, the increase in fluidity was not entirely governed by water content, as the binder composition and alkali dosage also affected the rheological behavior of the paste. The incorporation of LP improved particle packing and reduced friction between particles, while moderate SS content contributed to a more favorable particle-size distribution. In contrast, excessive SS replacement reduced the content of highly reactive GGBS and altered the balance between particle packing and hydration kinetics. Consequently, although some mixtures exhibited high fluidity, this was not always associated with desirable overall performance.

For setting behavior, the solid-waste composition exhibited a stronger influence than the water-to-binder ratio. Mixtures containing higher GGBS contents generally showed shorter setting times because the highly reactive glassy phases in GGBS dissolved rapidly under alkaline conditions, accelerating the formation of early hydration products. In contrast, increasing SS replacement significantly prolonged setting time due to its lower early-age reactivity and slower dissolution kinetics. For example, Z7, containing the highest SS

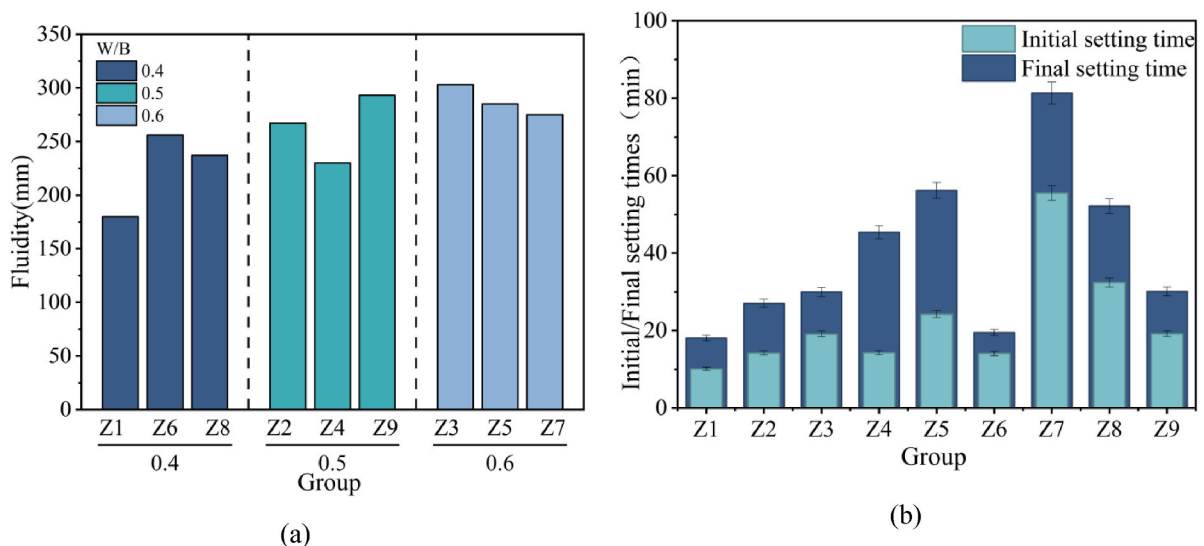


Fig. 3. Fluidity and setting time of the multiple-waste systems: (a) fluidity; (b) setting times.

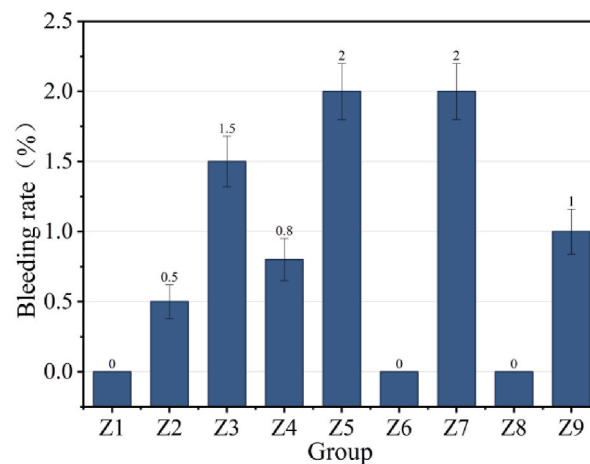


Fig. 4. Variation in 2h bleeding rate of the multiple-waste paste.

Table 4

Range analysis results of workability indices in the orthogonal experiments.

Evaluation index	Range R	Solid-waste proportion (A)	Alkali dosage (B)	W/B (C)	Order of influence
Fluidity (mm)	128.3	21.6	42.7	64.0	C > B > A
Setting time (min)	18.7	9.2	5.8	3.7	A > B > C
Bleeding rate	1.82	0.57	0.41	0.84	C > A > B

content (35%) and lowest GGBS content (29%), exhibited the longest setting time among all mixtures. Although increasing the alkali dosage partially accelerated precursor dissolution, its effect was less significant than that of the binder composition.

Among all mixtures, Z6 exhibited the most balanced fresh performance, with a fluidity of approximately 256 mm, initial and final setting times of 14.1 and 19.5 min, respectively and zero bleeding after 2 h. Although some mixtures showed either higher fluidity or longer setting times, excessive flowability may compromise stability, while excessively long setting may delay hardening and reduce construction efficiency. Therefore, Z6 was selected because it provided an appropriate balance between workability, setting behavior, and fresh-state stability, which is essential for practical applications. These results suggest that the fresh performance of the multi-waste system depends on the balanced interaction among different components rather than the effect of any single constituent.

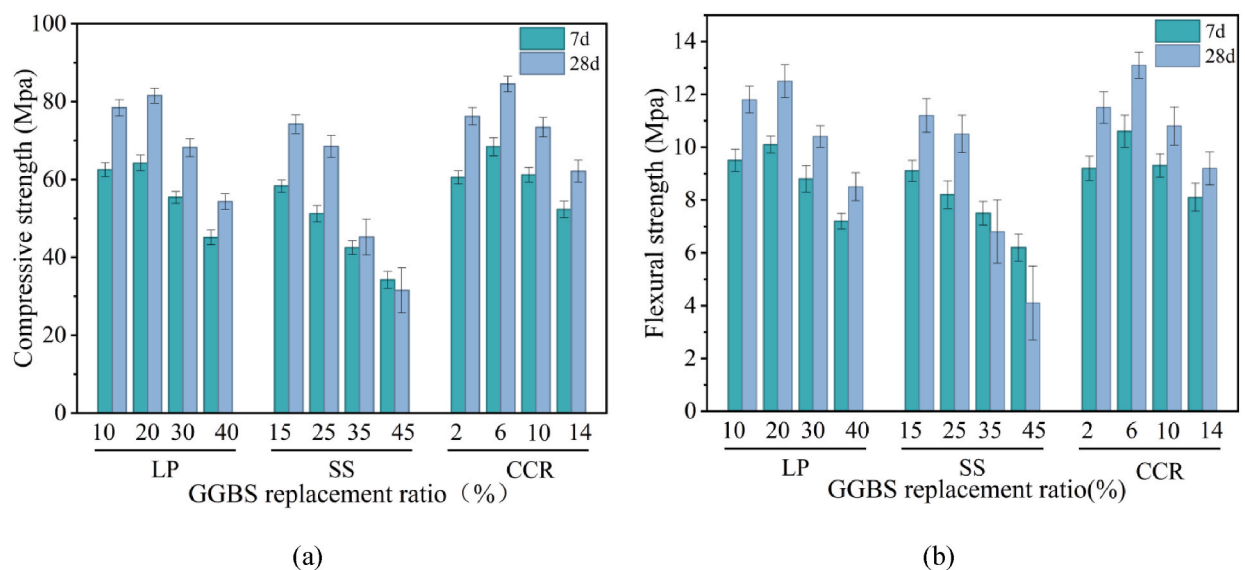


Fig. 5. Mechanical properties of alkali-activated pastes in the single-factor binary systems: (a) compressive strength; (b) flexural strength.

3.2. Mechanical properties

Mechanical strength reflects the combined effects of reaction-product formation, pore-structure densification, and potential microstructural defects [50]. As illustrated in Fig. 5, the mechanical performance of the mixtures was jointly controlled by LP, SS, and CCR through their respective roles in matrix densification, reactive-phase balance, and internal alkaline activation. Fig. 5(a) shows the compressive strength development of all solid-waste binder mixtures. The incorporation of LP exhibited an optimum dosage range. When the LP content increased from 10% to 20%, the 28-day compressive strength increased from 78.4 MPa to 81.5 MPa. This improvement indicates that appropriate LP content promotes the precipitation of C-(A)-S-H gels through heterogeneous nucleation and enhances matrix densification via a micro-filling effect. However, further increasing LP content diluted the reactive GGBS fraction and reduced the availability of dissolved Si and Al species, leading to a reduction in both early- and later-age compressive strength. Similarly, CCR showed a non-linear influence on compressive strength. The CCR6 mixture achieved the highest compressive strength at both 7 and 28 days, suggesting that an optimal CCR dosage provides sufficient alkalinity and additional calcium ions to accelerate GGBS depolymerization and gel formation. In contrast, excessive CCR incorporation may lead to rapid precipitation of hydration products, which hinders ion diffusion and limits further reaction progress [51]. In contrast, SS incorporation generally reduced compressive strength. The SS45 mixture exhibited a decrease from 34.2 MPa at 7 days to 31.5 MPa at 28 days, indicating weak long-term strength development. This behavior is mainly attributed to the low intrinsic reactivity of SS and the presence of inert or slowly reacting crystalline phases, including free CaO and MgO, which do not significantly contribute to the formation of binding gels [52] (see Fig. 6).

The flexural strength showed a similar but more sensitivity-driven trend compared to compressive strength, as it is more dependent on matrix continuity and microcrack resistance. For SS mixtures, flexural strength decreased from 6.2 MPa at 7 days to 4.1 MPa at 28 days, confirming the detrimental effect of SS on matrix cohesion and crack-bridging ability. This reduction is associated with a less compact microstructure and weaker gel connectivity. In contrast, LP and CCR exhibited trends consistent with compressive strength results, where moderate dosages improved flexural performance due to enhanced matrix densification and improved interfacial bonding. The optimum dosage of LP and CCR are 20% and 6%, respectively. However, excessive replacement levels led to a reduction in flexural strength due to increased heterogeneity and microstructural discontinuities.

The orthogonal experiments revealed that the strength development of the multiple-waste system was jointly governed by the proportions of GGBS, SS, LP, and CCR. Among the orthogonal mixtures, Z6 demonstrated superior and continuously increasing strength. At 7 days, Z6 achieved a compressive strength of 52.6 MPa and a flexural strength of 10.2 MPa, which further increased to 75.8 MPa and 12.1 MPa at 28 days, respectively. This continuous strength gain indicates that moderate incorporation of SS and LP does not hinder the sustained reaction capacity of the system. In the alkaline environment provided by CCR, GGBS and part of the SS were progressively activated, while LP contributed to matrix densification through nucleation and micro-filling effects. The synergistic interaction among these components enabled Z6 to maintain a balanced development of both early- and later-age mechanical properties. In contrast, Z8 exhibited an opposite later-age strength evolution. Although it reached relatively high 7-day strengths of 61.43 MPa (compressive) and 11.86 MPa (flexural), these values significantly declined to 41.40 MPa and 5.13 MPa at 28 days, respectively. Compared with Z6, Z8 contained higher SS and LP contents but lower GGBS content under the same water-to-binder ratio, indicating that excessive SS and LP disrupted the balance among reactive-phase supply, internal alkaline activation, and structural stability. As a result, the later-age strength regression of Z8 is therefore not attributable primarily to W/B ratio, but to insufficient reactive precursors, accumulated steel-slag residual crystalline phases, and delayed transformation of f-CaO-bearing high-calcium

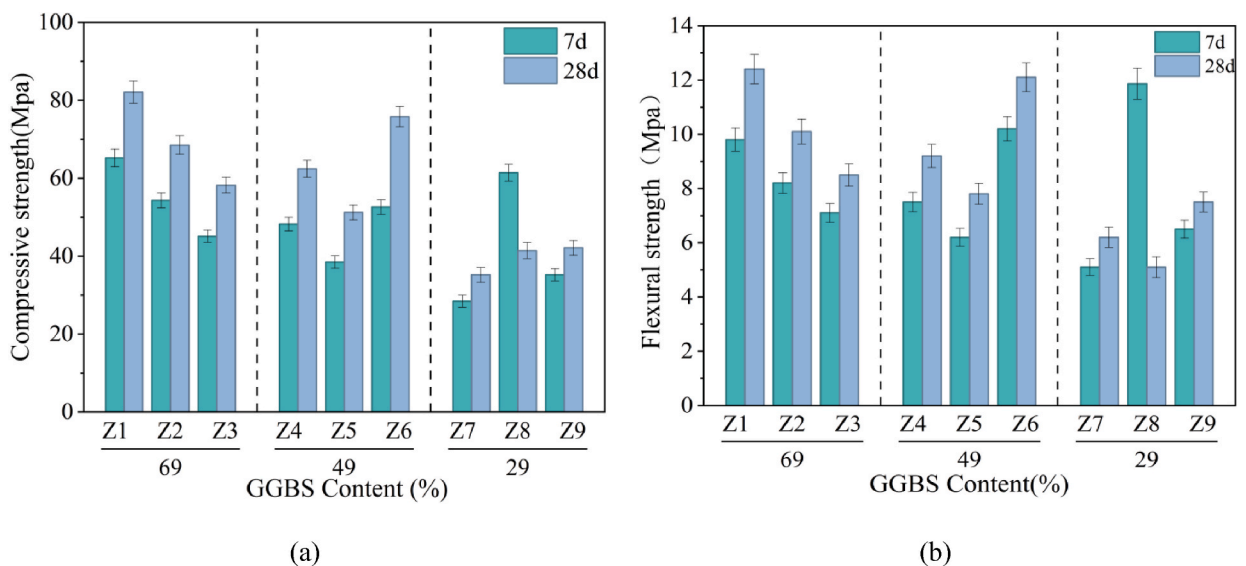


Fig. 6. Mechanical properties of alkali-activated pastes in the orthogonal multiple-waste systems: (a) compressive strength; (b) flexural strength.

phases [53]. The optimum mixture was selected based on a comprehensive evaluation of both fresh-state and hardened properties rather than a single performance indicator. The evaluation criteria included fluidity, initial and final setting times, bleeding rate, compressive strength, and flexural strength. Among the investigated mixtures, Z6 exhibited the most balanced overall performance, combining high fluidity, appropriate setting behavior, zero bleeding, and the highest mechanical strengths. Therefore, Z6 was identified as the optimum mixture and selected for subsequent microstructural characterization.

3.3. Microstructural analysis

3.3.1. X-ray diffraction

X-ray diffraction (XRD) was employed to characterize crystalline phases and track residual-phase evolution in different mixtures at 7 and 28 d. LP10 and SS45 were selected as reference mixtures to isolate the individual contributions of LP and SS, respectively. Since C-(A)-S-H gels are predominantly poorly crystalline, XRD was mainly used to identify crystalline residual phases and compare their relative signal intensities rather than quantify gel content [54].

As shown in Fig. 7, All samples exhibited characteristic calcite reflections at both ages, with dominant peaks at 2θ 29.5°. In addition, a broad diffuse hump was observed within the 25–35° range, indicating the presence of poorly crystalline hydration products. LP10 displayed the most intense calcite peaks, confirming that LP remained predominantly as crystalline CaCO_3 in the alkali-activated system, where it provides physical filling and heterogeneous nucleation sites. The single-factor reference mixtures clarified the origins of the residual crystalline phases. LP10 exhibited only calcite reflections superimposed on a diffuse background and showed no discernible residual peaks in the characteristic steel-slag regions. In contrast, SS45 exhibited distinct peaks near 31–32°, 42–43°, and 60–65°, corresponding to the RO phase, dicalcium silicate (C_2S), and other characteristic steel-slag residual phases. Additional signals near 37.3° and 53.8° were associated with f-CaO-bearing high-calcium residual phases. These observations confirm that the residual crystalline signals observed in the quaternary mixtures originated primarily from SS rather than LP. SS45 exhibited more pronounced CH-related reflections, suggesting that higher portlandite content is retained or formed in systems with elevated SS proportions — consistent with the higher CH-related mass losses observed for Z8 in TG–DSC analysis (Section 3.3.2).

The phase evolution of Z6 and Z8 revealed marked differences. Z8 exhibited stronger calcite reflections than Z6, consistent with its higher LP content. Characteristic diffraction peaks of portlandite (CH) were detected at $2\theta \approx 18.1^\circ$, 34.1° , and 50.8° . However, these peaks remained relatively weak in both mixtures and showed little increase with curing age, indicating that the $\text{Ca}(\text{OH})_2$ introduced by CCR was largely consumed during GGBS activation and partial SS hydration rather than accumulating as a crystalline phase. Compared with Z6, Z8 exhibited more pronounced residual steel-slag peaks and stronger signals associated with f-CaO-bearing high-calcium phases. Furthermore, the peaks near 37.3° and 53.8° weakened from 7 d to 28 d, suggesting partial later-age transformation of these high-calcium residual phases [55]. In contrast, the diffraction pattern of Z6 remained relatively stable throughout curing. Although distinct calcite reflections were present, neither CH-related peaks nor steel-slag residual peaks increased appreciably between 7 d and 28 d. These observations indicate a relatively stable reaction process in which the alkalinity supplied by CCR, the nucleation and filling effects of LP, and the high reactivity of GGBS acted synergistically to sustain hydration. Consequently, the later-age strength development of Z6 can be attributed primarily to the continued formation of poorly crystalline C-(A)-S-H gels and progressive matrix densification rather than accumulation of crystalline products. By comparison, the stronger residual steel-slag signals and more pronounced high-calcium phase evolution observed in Z8 indicate a more heterogeneous reaction process under elevated SS incorporation. These phase-level differences provide evidence that the later-age strength deterioration of Z8 was

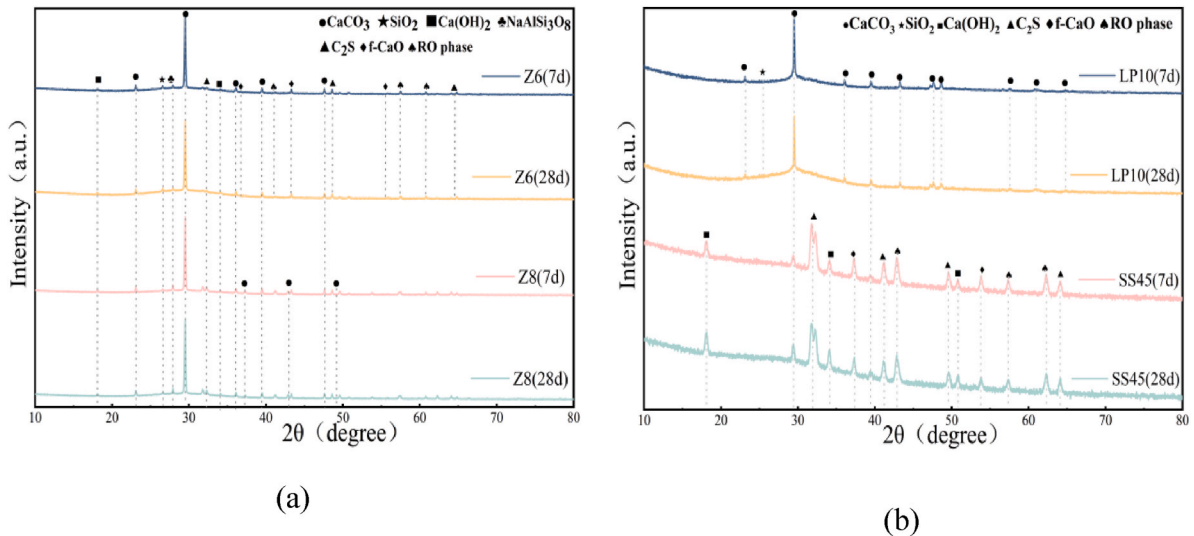
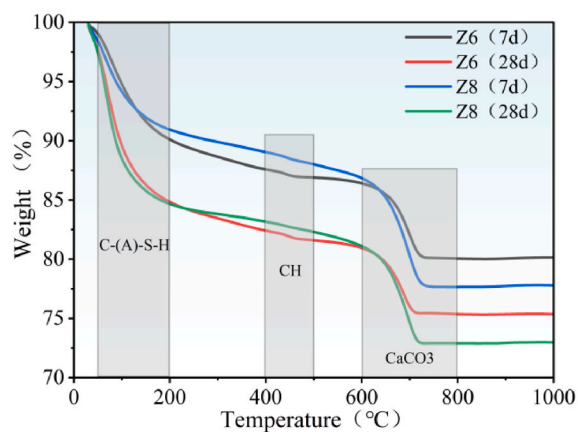
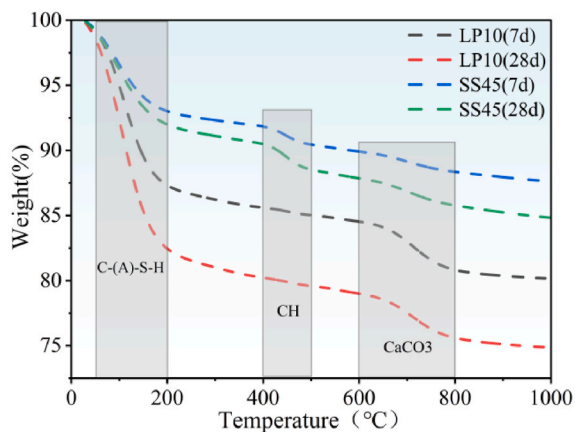


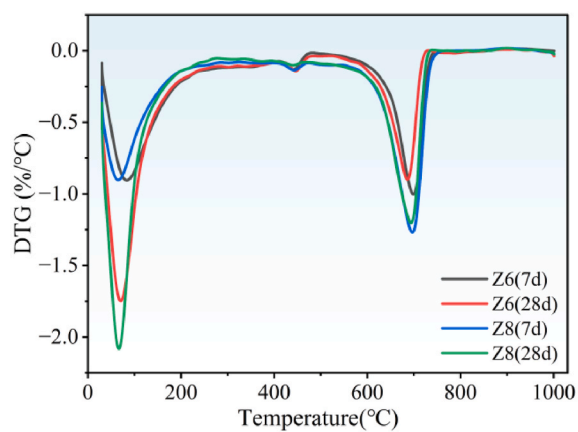
Fig. 7. XRD patterns of alkali-activated pastes at 7d and 28d: (a) Z6 and Z8; (b) LP10 and SS45.



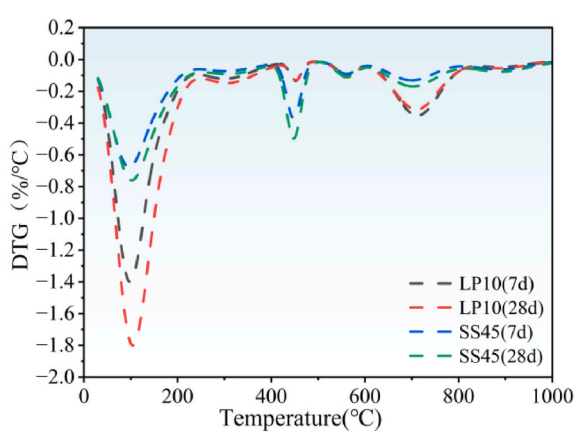
(a)



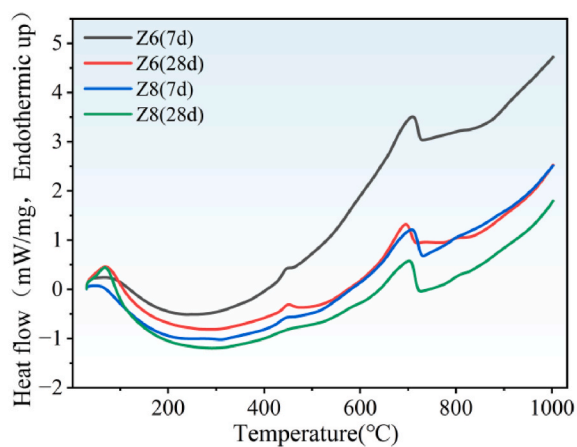
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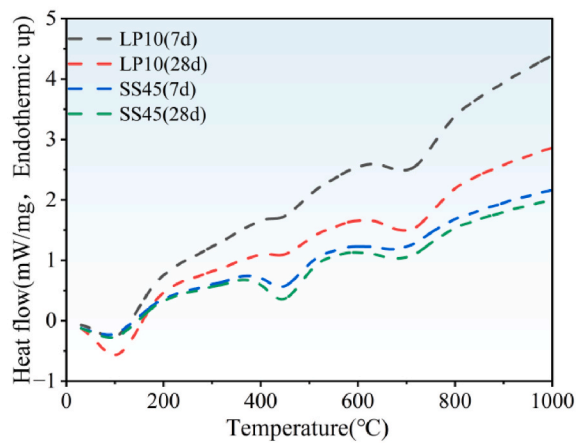
(c)



(d)



(e)



(f)

Fig. 8. TG–DTG/DSC curves of Z6, Z8, and the single-factor reference groups LP10 and SS45: (a) TG curves of Z6 and Z8; (b) TG curves of LP10 and SS45; (c) DTG curves of Z6 and Z8; (d) DTG curves of LP10 and SS45; (e) DSC curves of Z6 and Z8; (f) DSC curves of LP10 and SS45.

associated with the accumulation of residual crystalline phases and the transformation of f-CaO-bearing high-calcium phases [56].

Overall, the XRD results revealed clear phase-level differences between the optimized mixture (Z6) and the mixture exhibiting later-age strength deterioration (Z8). While Z6 maintained a relatively stable hydration process dominated by the formation of poorly crystalline products, Z8 contained larger amounts of residual crystalline phases and exhibited more pronounced high-calcium phase evolution. Since XRD cannot effectively quantify poorly crystalline hydration products, TG–DSC analysis was subsequently employed to further characterize hydration-product formation and residual CH content.

3.3.2. TG–DTG analysis

TG–DTG was employed to characterize hydration-product formation, residual portlandite (CH) content, and the evolution of carbonate-related phases. Fig. 8 presents the TG, DTG, and DSC curves of the alkali-activated pastes at 7 and 28 d, showing good agreement with the XRD results. All mixtures exhibited three characteristic thermal peaks. The first peak, located at approximately 80 °C, was mainly associated with the dehydration of C-(A)-S-H gels and other low-temperature hydration products. The second peak, centered near 430 °C, corresponded to the dehydroxylation of portlandite (CH). The third peak, appearing at approximately 710 °C, was attributed to the decomposition of CaCO_3 and related carbonate phases [57–59]. The single-factor reference mixtures revealed distinct differences in hydration-product development. The low-temperature mass loss of LP10 increased noticeably with curing age, indicating continuous formation of hydration products in the absence of elevated SS content. In contrast, SS45 exhibited a relatively weaker low-temperature response, suggesting that replacing GGBS with a high proportion of SS reduced the availability of reactive glassy phases and consequently limited hydration-product accumulation [39,58]. Meanwhile, SS45 showed a more pronounced CH-related thermal response than LP10, indicating greater retention of calcium-bearing phases in the high-steel-slag system. These observations are consistent with the residual crystalline phases identified by XRD. The quaternary mixtures exhibited markedly different hydration evolutions. For Z6, the low-temperature mass loss increased from 12.30% at 7 d to 17.45% at 28 d, indicating continuous hydration-product formation during curing. A similar increase was observed for Z8, whose low-temperature mass loss increased from 10.21% to 16.20%. These results suggest that hydration continued in both mixtures up to 28 d. Therefore, the later-age strength reduction of Z8 cannot be attributed to hydration stagnation or insufficient hydration-product formation, but rather suggests that the newly formed products may not have effectively contributed to the development of a dense and mechanically efficient matrix structure.

More significant differences were observed in the evolution of calcium-bearing phases. The CH-related mass loss of Z6 increased only slightly from 0.67% at 7 d to 0.76% at 28 d, whereas Z8 exhibited substantially higher values, decreasing from 1.49% to 1.23% while remaining consistently above those of Z6. Combined with the stronger residual steel-slag signals and f-CaO-associated diffraction peaks identified by XRD, these results suggest that elevated SS incorporation promoted the accumulation of residual calcium-bearing phases and increased the likelihood of later-age transformation of high-calcium residual phases [59]. The mass loss in the 600–800 °C range was mainly associated with decomposition of carbonate-related phases. Z8 consistently exhibited higher mass loss than Z6 in this temperature range, which corresponds to its higher LP content and greater proportion of carbonate phases. The distinct endothermic

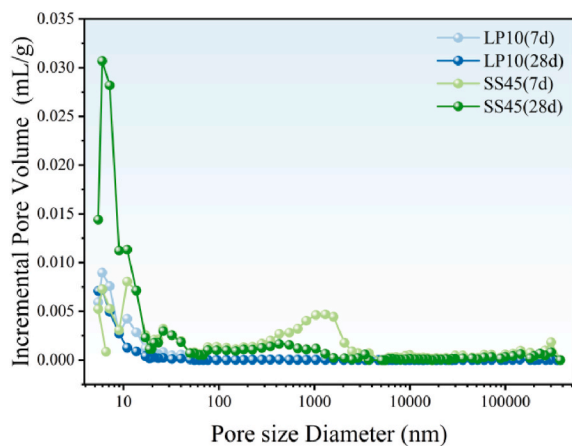
peak observed at approximately 693–704 °C in the DSC curves further confirmed that calcite decomposition was the dominant thermal event within this region [58,60]. Therefore, the mass loss within this range primarily reflects carbonate content and should not be interpreted as a direct indicator of hydration-product formation.

Overall, the TG–DSC results demonstrate that hydration products continued to form in both Z6 and Z8 throughout curing. However, despite comparable increases in low-temperature hydration-product-related mass loss, Z8 retained a stronger CH-related response and exhibited more pronounced calcium-bearing phase evolution than Z6. Combined with the XRD results, these findings indicate that the later-age strength reduction of Z8 was not associated with insufficient hydration but was more closely associated with heterogeneous phase evolution and the persistence of residual calcium-bearing phases in the high-SS system. Further verification of whether these products effectively improved pore refinement and matrix compactness is provided through MIP analysis.

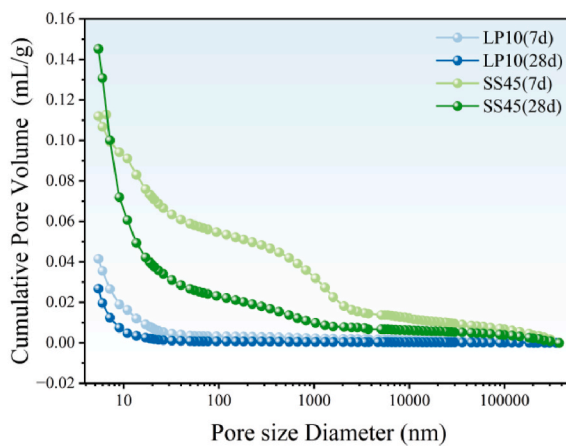
3.3.3. Pore structure analysis

MIP was employed to verify whether the continuously formed hydration products effectively contributed to pore refinement and matrix densification. Based on the integrated XRD and TG–DSC results, four representative mixtures were selected for detailed pore structure analysis: LP10 as the steel-slag-free control mixture, SS45 as the high-steel-slag single-factor mixture, Z6 as the optimized mixture, and Z8 as the mixture exhibiting later-age strength reduction. All mixtures were analyzed at 7d and 28d. Fig. 9 present the outcome of the MIP test performed on the alkali activated paste sample, which serves to assess the pore structure. Fig. 9(a and b) illustrates the distribution of the pore size and the cumulative pore volume of LP10 and SS45, respectively. Fig. 9(c and d) illustrates the distribution of the pore size and the cumulative pore volume of Z6 and Z8, respectively. Fig. 9(e) illustrates the median and average pore diameters of the samples, while Fig. 9(f) indicates the porosity and pore volume fraction of the samples. According to prior research [61], concrete pore types can be classified into gel pores (below 10 nm), mesopores (10–100 nm), capillary pores (100–10000 nm), and macro pores (above 10000 nm) based on pore size and their impact on the material's matrix structure.

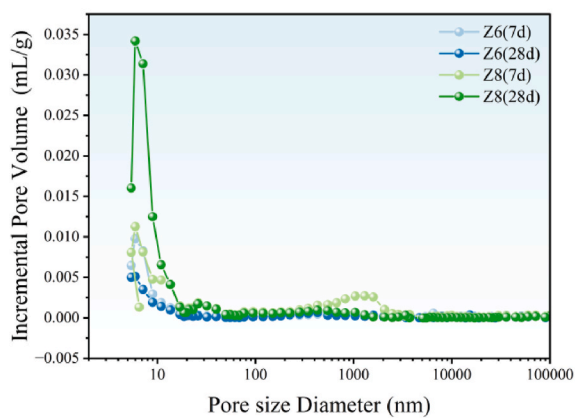
The evolution of pore structure in the single-factor reference mixtures revealed distinctly different behaviors. For LP10, the total intrusion volume decreased from 0.0415 mL/g at 7 d to 0.0268 mL/g at 28 d, accompanied by a reduction in the volume fraction of larger accessible pore volume. These results indicate that hydration in the steel-slag-free system continuously refined the pore structure and promoted matrix densification. In contrast, SS45 showed a substantial increase in total intrusion volume from 0.1128 to 0.1452 mL/g, while the proportion of pore throats smaller than 10 nm increased sharply from 19.3% to 58.2%. Combined with the residual crystalline phases and CH-related responses identified by XRD and TG–DSC, these results indicate that excessive SS incorporation increased the likelihood of pore-structure instability during later curing. The quaternary mixtures exhibited markedly



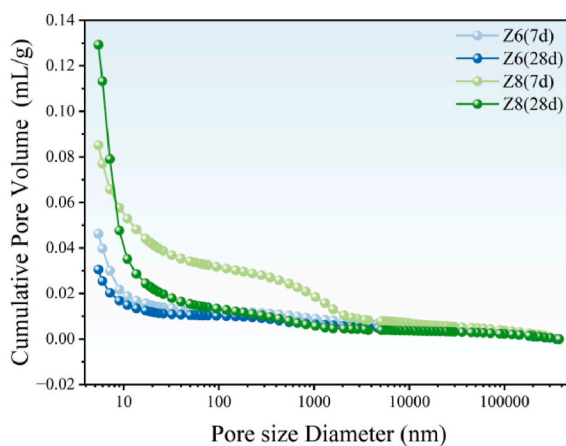
(a)



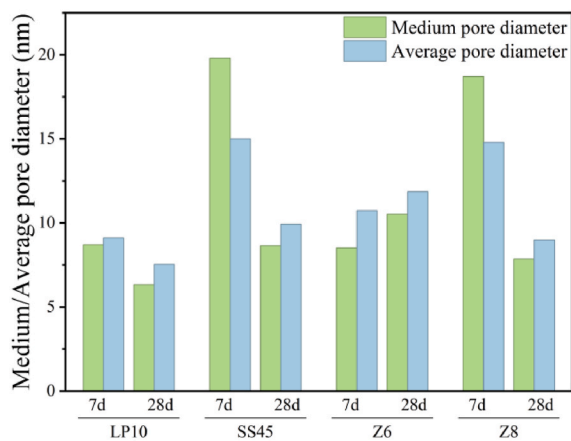
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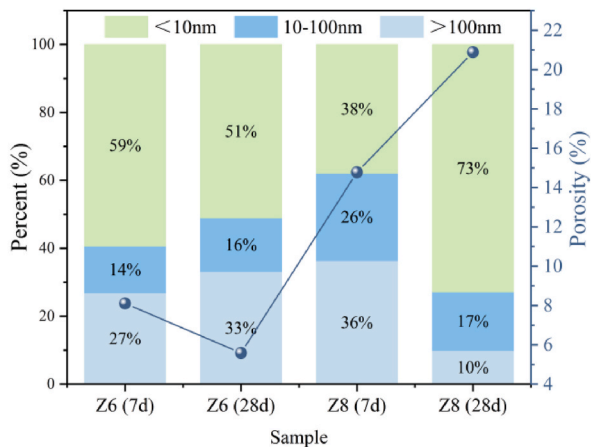
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(d)



(e)



(f)

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Fig. 9. MIP pore-structure evolution of different systems: (a) Pore size distribution of LP10 and SS45; (b) cumulative pore volume of LP10 and SS45; (c) Pore size distribution of Z6 and Z8; (d) cumulative pore volume of Z6 and Z8; (e) Medium and average pore diameter; (f) Porosity and pore volume fraction.

different pore-evolution characteristics. Z6 underwent progressive and continuous pore-structure densification during curing. Total mercury intrusion volume decreased from 0.0462 to 0.0305 mL/g, total pore surface area from 17.24 to 10.29 m²/g, and porosity from 8.10% to 5.58% between 7 d and 28 d. Simultaneously, intrinsic permeability decreased and tortuosity increased, indicating that poorly crystalline hydration products effectively filled existing pores and reduced pore connectivity. These changes are consistent with the continuous development of mechanical strength in Z6 [62]. In contrast, Z8 exhibited substantial later-age pore-structure deterioration. The total intrusion volume increased from 0.0852 to 0.1293 mL/g, while the total pore surface area and porosity increased from 23.04 to 57.58 m²/g and from 14.78% to 20.88%, respectively. The simultaneous increase in these parameters indicates that the matrix did not undergo progressive densification during curing but instead developed a more complex and interconnected accessible pore network. The similarity between the pore-evolution trends of Z8 and SS45 further suggests that the deterioration of Z8 was primarily associated with elevated SS incorporation and the resulting accumulation of residual crystalline phases rather than the LP content alone [63].

The pore-size distribution of Z8 provides further insight into this apparently contradictory evolution behavior. The volume of fine pores (<10 nm) increased from 0.0323 mL/g to 0.0941 mL/g, while its volume fraction increased from 37.92% to 72.86%. Conversely, the volume of coarse pore (>100 nm) decreased from 0.0342 mL/g to 0.0155 mL/g over the same period. In parallel, the median pore diameter (D₅₀) and average pore diameter decreased from 18.71 to 14.79 nm at 7 d to 7.85 and 8.98 nm at 28 d, respectively. This apparent reduction should not be interpreted as pore refinement. Instead, it reflects the ink-bottle effect inherent to MIP measurements. When large pore bodies, microcracks, or interfacial defects are connected to the external surface through narrow entrance throats, the measured pore size is governed by the throat diameter rather than the actual internal void dimension, leading to an apparent shift toward smaller pore sizes [64]. The MIP results provide pore-structure evidence for both the interactive benefits and the limitations of the multiple-waste system. In Z6, the combined effects of LP, SS, CCR, and GGBS promoted progressive pore refinement, reduced pore interconnectivity, and enhanced matrix compactness. In contrast, excessive SS incorporation in Z8 shifted the system from synergistic densification toward pore-network instability. Although hydration products continued to form, their contribution to effective matrix densification was limited, resulting in the development of a highly complex accessible pore network. Since MIP cannot directly characterize particle–matrix interfaces or microcrack morphology, SEM–EDS analysis was subsequently employed to further investigate the microstructural differences between Z6 and Z8.

It should be noted that the pore size distribution obtained from MIP represents the pore-throat size rather than the actual pore-body size. In addition, the ink-bottle effect may lead to an underestimation or overestimation of certain pore classes. Therefore, the MIP results should be interpreted as comparative indicators of pore structure evolution rather than absolute pore-size measurements.

3.3.4. SEM–EDS analysis

SEM–EDS analysis of fracture surfaces was conducted at 7 d and 28 d to characterize matrix continuity, particle–matrix interfacial bonding, and microcrack development in Z6 and Z8, and to further verify the pore-structure evolution inferred from the MIP results. The corresponding results are presented in Figs. 10–13. At 7 d, Z6 had already developed a relatively continuous and homogeneous gel-rich matrix. EDS mapping of representative regions showed dominant enrichment in Al (15.37 wt%), Si (54.13 wt%), and Ca (30.50 wt %), consistent with the early formation of Ca–Si–Al-rich C-(A)-S-H-type products [65]. No obvious cracks or interfacial defects were observed, and the interparticle spaces were largely occupied by hydration products. This morphology suggests that the early

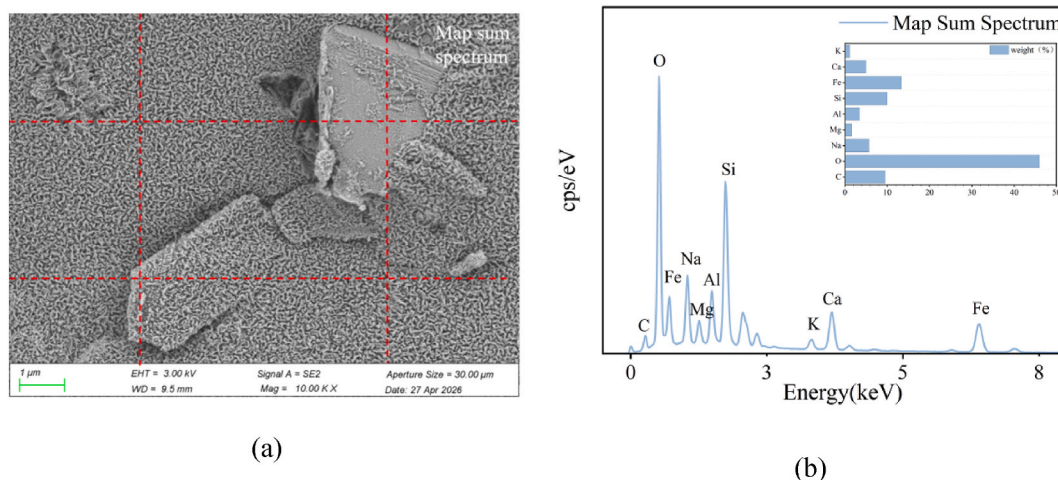


Fig. 10. SEM morphology of Z6 at 7d.

mechanical performance of Z6 was associated with rapid hydration-product formation and effective matrix consolidation. At 28 d, the fracture surface of Z6 exhibited a denser and more compact gel-rich matrix, with particles tightly bonded to the surrounding matrix and no discernible interfacial debonding or significant cracking. These observations are consistent with the MIP results, which showed simultaneous reductions in total intrusion volume, pore surface area, and porosity, indicating progressive matrix densification during curing.

In contrast, Z8 exhibited a partially consolidated and heterogeneous matrix at 7 d. Fracture surfaces contained abundant residual particles, localized void regions, and loosely packed gel zones, indicating markedly inferior microstructural homogeneity compared to Z6. EDS point analyses identified discrete regions with elevated Fe concentrations, consistent with the presence of unreacted steel-slag particles or Fe-rich crystalline residual phases. At 28 d, particle–matrix interfacial debonding and localized microcracks became prominent features of Z8 fracture surfaces. These defects were concentrated in the vicinity of residual particles and propagated along interparticle boundaries and through gel-deficient zones. The observed morphology suggests that elevated SS incorporation and reduced reactive GGBS contribution increased microstructural heterogeneity and promoted localized defect accumulation during later curing. The progressive accumulation of these interfacial defects directly explains the 28-day strength regression of Z8 despite continued hydration-product formation.

The SEM–EDS observations provide direct microstructural evidence supporting the mechanisms inferred from the XRD, TG–DSC, and MIP analyses. In Z6, the continuous gel-rich matrix and well-bonded particle–matrix interfaces are consistent with progressive matrix densification and sustained strength development during curing. In contrast, the residual-particle debonding, localized microcracking, and porous gel-deficient regions observed in Z8 correspond well with the MIP evidence of increased pore surface area and a more complex accessible pore network, as well as the XRD identification of abundant residual steel-slag crystalline phases. Together, these results indicate that appropriate proportioning of LP, SS, CCR, and GGBS enables continuous hydration to be effectively translated into matrix densification and structural stability. Conversely, excessive SS incorporation combined with insufficient reactive GGBS supply promotes microstructural heterogeneity, interfacial deterioration, and pore-network instability, ultimately leading to the loss of later-age mechanical performance.

The SEM–EDS observations presented in this study provide qualitative evidence of the microstructural evolution and are intended to support the findings obtained from XRD, TG–DSC, and MIP analyses. Although the observed matrix densification and reduction in microcracks are consistent with the improved mechanical properties, quantitative image analysis and statistical evaluation of EDS data were beyond the scope of the present study. These aspects will be investigated in future work to provide a more comprehensive assessment of the microstructural characteristics.

The microstructural evidence presented above establishes strong correlations between phase assemblage, pore structure, and mechanical performance. However, several limitations should be acknowledged. First, XRD analysis in this study is semi-quantitative; definitive phase quantification would require Rietveld refinement or internal standard methods. Second, the hypothesized role of f-CaO-bearing phase transformation in Z8 strength regression is inferred from XRD peak evolution and TG–DSC mass losses, but direct measurement of free CaO and free MgO content would provide more direct evidence. Third, expansion measurements are needed to confirm whether the observed microstructural deterioration translates to macroscopic volume instability. These additional investigations are recommended as future work to establish a definitive causal chain.

4. Conclusions

This study developed a GGBS–SS–LP–CCR quaternary alkali-activated paste and systematically investigated the effects of multi-source solid-waste composition on fresh properties, mechanical performance, and microstructural evolution. The following principal conclusions are drawn.

- (1) The improved fresh properties of the quaternary solid-waste system resulted from the functional complementarity among the different components. LP improved particle packing and bleeding resistance, SS enhanced initial fluidity and prolonged setting behavior, while CCR provided internal alkalinity to accelerate GGBS activation. The optimized mixture (Z6) achieved fluidity of 256 mm, initial and final setting times of 14.1 and 19.5 min, respectively, together with a 2 h bleeding rate of 0, demonstrating balanced fresh-state performance suitable for applications requiring controlled setting and minimal bleeding.
- (2) The sustained strength development of Z6 was associated with a synergistic densification mechanism involving reactive-phase supply, internal alkaline activation, and filling–nucleation effects. The combined action of GGBS, CCR, LP, and appropriate SS incorporation promoted continuous formation of poorly crystalline C-(A)-S-H gels, matrix densification, and improved particle–matrix bonding. Consequently, the compressive strength increased from 52.6 MPa at 7 d to 75.8 MPa at 28 d, while the flexural strength increased from 10.2 MPa to 12.1 MPa.
- (3) Excessive SS and LP incorporation combined with insufficient GGBS supply shifted the system from synergistic densification toward structural instability. Although hydration products continued to form in Z8, abundant residual crystalline phases, heterogeneous phase evolution, and progressive pore-network complexity reduced the effectiveness of later-age matrix densification. This behavior ultimately resulted in microstructural deterioration and later-age strength reduction. These findings indicate that the performance of the multiple-waste system strongly depends on maintaining an appropriate synergistic component ratio.

Overall, the results demonstrate that rational proportioning of multiple solid wastes can effectively achieve both desirable engineering performance and microstructural stability, providing a promising approach for applications requiring controlled setting and

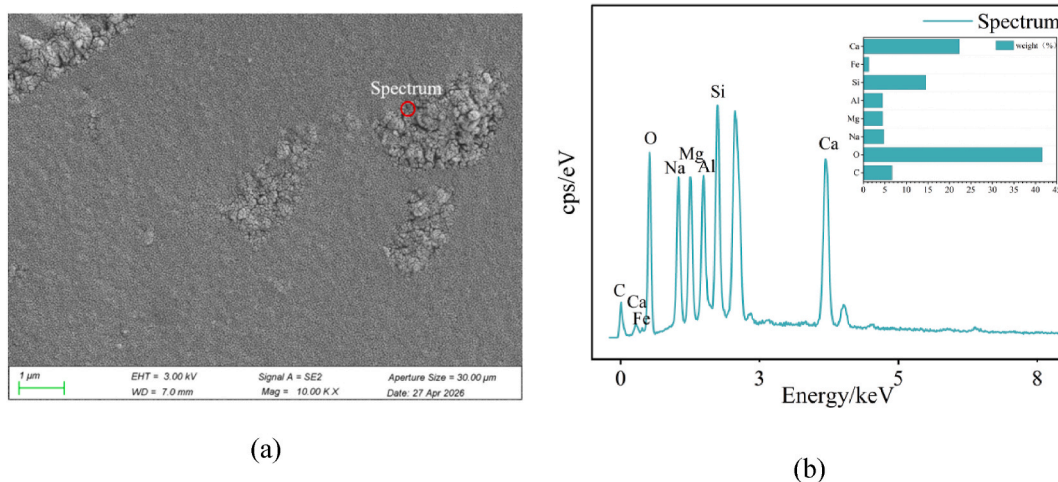


Fig. 11. SEM morphology of Z6 at 28d.

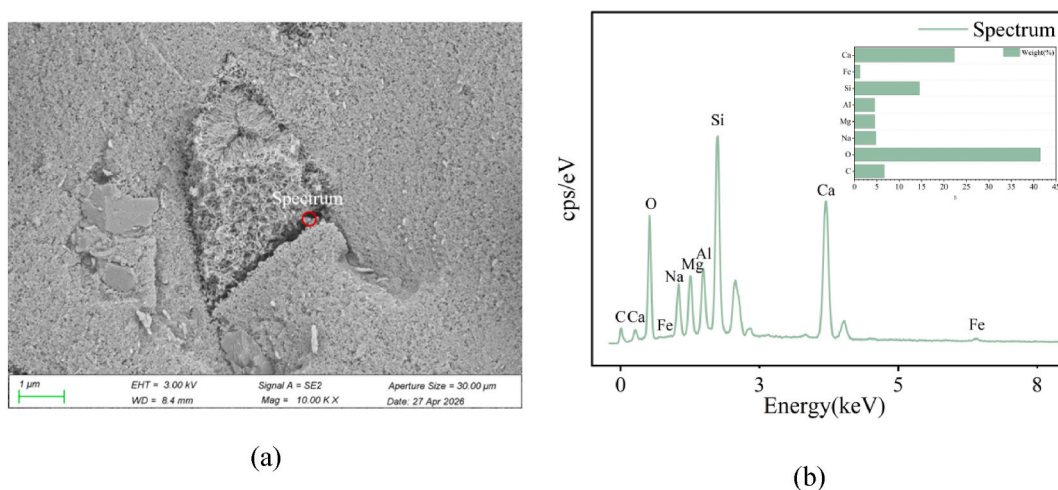


Fig. 12. SEM morphology of Z8 at 7d.

minimal bleeding.

5. Limitations and future studies

The present study focused on optimizing the fresh properties, mechanical performance, and microstructural evolution of a quaternary alkali-activated binder through single-factor and orthogonal experimental designs. Although the proposed approach successfully identified an optimum mixture, more comprehensive experimental methods, such as full factorial or response-surface designs, are recommended to quantitatively evaluate the individual and interactive effects of each constituent and establish statistically robust optimization models. The experimental program was limited to 28 days of curing. Therefore, the long-term dimensional stability and durability of the proposed binder, including drying shrinkage, autogenous shrinkage, expansion associated with residual f-CaO/f-MgO, water resistance, and sulfate/chloride resistance, should be systematically investigated to verify its long-term engineering performance.

Although the developed binder exhibited favorable fresh-state properties and mechanical strength, its practical applicability in underground grouting requires further validation through rheological characterization, flow retention, pumpability, injectability, permeability reduction, and durability under representative service environments. The microstructural interpretation presented in this study is primarily based on qualitative analyses using XRD, TG-DSC, MIP, and SEM-EDS. Future investigations should incorporate quantitative image analysis, statistical evaluation of EDS measurements, and interfacial transition zone characterization to provide a more comprehensive understanding of the strengthening mechanisms.

Finally, while the extensive utilization of industrial by-products demonstrates considerable potential for reducing Portland cement

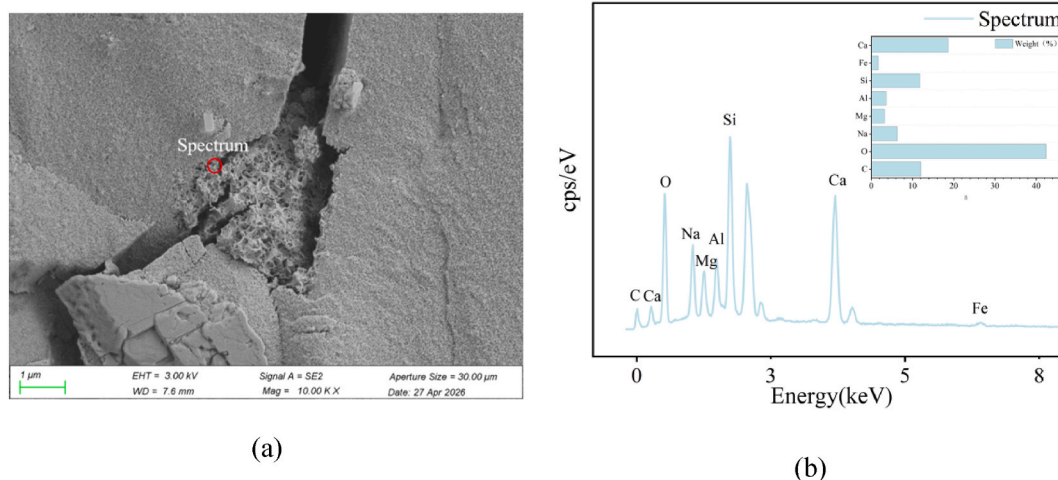


Fig. 13. SEM morphology of Z8 at 28d.

consumption and promoting resource efficiency, comprehensive sustainability assessments, including life-cycle assessment, embodied carbon analysis, economic evaluation, and heavy-metal leaching tests, are required to fully quantify the environmental benefits of the proposed binder.

CRediT authorship contribution statement

Jinxin Zhang: Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Mahmoud Elmaasrawy:** Supervision, Writing – review & editing. **Ming Zhou:** Supervision, Visualization. **Shihao Zhao:** Investigation, Methodology, Software. **Baoyuan Li:** Supervision, Visualization. **Jing Peng:** Supervision, Visualization. **Xu Gao:** Supervision, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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